

SEPEHR ZHAND, Ph.D.

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Senior Geospatial Data Scientist | Urban & Transport Systems

Ph.D. specialized in the design of scalable analytical frameworks for complex urban networks. Proven track record in architecting production-grade data pipelines on **Google Cloud Platform** and translating high-fidelity spatial modeling into operational decision-support tools. Experienced in modeling systems exceeding **1 million network nodes** and delivering validated digital solutions adopted by municipal authorities for long-term infrastructure and transport planning.

TECHNICAL SKILLS

Programming:	Python (NumPy, Pandas, SciPy, scikit-learn, NetworkX, OSMnx, Fiona, Rasterio), R, SQL,.
Data Engineering:	Multi-source Data Integration, Automated ETL Pipelines, Google Cloud Platform, PostgreSQL/PostGIS.
Reproducibility:	Git/GitHub (Version Control), Data Validation, Reproducible Research.
Spatial Analytics:	Agent-Based Modeling (DepthmapX, Net Logo), Network Topology, Spatial Statistics.
Visualization:	Interactive Dashboards (Mapbox GL JS, JavaScript), Matplotlib, QGIS

PROFESSIONAL EXPERIENCE

Associate | Geospatial data Scientist *Jan 2020 – Present*

Studio Bednarski Ltd., London, UK

- Lead technical data strategy for multidisciplinary urban design and transport projects, translating complex spatial diagnostics into operational insights for municipal stakeholders.
- Direct quantitative socio-spatial modeling for systems exceeding 1 million network nodes, integrating multi-source datasets (OSM, topography, demographic indicators) for infrastructure planning.
- Build reproducible geospatial pipelines using GeoPandas and PostGIS to generate accessibility metrics across complex urban systems.

Research Fellow | Spatial Data Scientist *March 2023 – December 2025*

University College London (TWIN2EXPAND), UK

- Architected and deployed automated, production-grade spatial data pipelines on GCP for 617 European urban systems using Python (OSMnx, Cityseer) and PostGIS.
- Ensured computational reproducibility by implementing Git-based version control and modular Python workflows for large-scale urban network analysis and multi-scale feature engineering.

- Produced structured datasets currently utilized in international urban analytics research and strategic policy applications.

Research Associate | Spatial Data Scientist *May 2021 – June 2023*

London School of Economics (LSE), UK

- Directed quantitative socio-spatial modeling across large-scale urban systems (~200k+ network nodes) to evaluate road connectivity and neighbourhood impacts.
- Built predictive and clustering models integrating land-use, deprivation indicators, and network structure datasets for policy research outputs.

Project Consultant | Geospatial Analyst *May – Sep 2019*

Space Syntax Ltd., London, UK

- Developed a bespoke land-value analytics engine for Melbourne by modeling a 500k+ node urban network and integrating multi-source spatial data.
- Produced an integrated value surface adopted by the municipality as a primary decision-support tool for strategic long-term planning.

ACADEMIC RESEARCH

Ph.D. Project | Urban and Spatial Analytics *Sep 2018 – Feb 2025*

UCL Space Syntax Lab, UK

- Developed a scalable spatial analytics framework for modeling non-linear urban dynamics across street-network systems exceeding 1M+ nodes.
- Built automated feature engineering workflows and applied machine learning methods to estimate missing socio-spatial attributes.
- Key Initiative: Developed the [Interactive Spatial Performance Monitor](#). Originally a byproduct of doctoral research, the tool was **adopted by the municipality** for real-time exploratory diagnostics and evidence-based planning.
- Designed the methodology to be simulation-ready, facilitating high-resolution scenario analysis for city-shaping decisions.

Key Project: Designed and deployed the [Tehran Spatial Disparity Index](#), a live interactive web mapping application built with Mapbox GL JS, visualising 13,848 H3 hexagons from a GCP pipeline with MLP-predicted deprivation scores and space syntax overlays.

EDUCATION

Ph.D. | Urban and Spatial Analytics **2025**

Space Syntax Laboratory, UCL, UK

MArch | Housing and Urbanism **2013**

AA School of Architecture, UK